Materials and Methods

Setting and participants. Data were collected from undergraduate students enrolled in a 10-week university course on virtual reality (VR) and AI over a 6-week period. At the beginning of the course, students were invited to participate in an Institutional Review Board-approved (IRB) study of how repeated exposure to VR and AI influenced their individual and group behavior. While all students who were enrolled in the course took part in all the course activities, only those who consented to participate in the study had their data included in the study. Safeguards implemented to ensure privacy and consent included review both by the IRB and a second university ethics organization, and third-party oversight of the consent process and data collection.

During the 6-week period, students were asked to turn in a "question and answer" analysis about the readings for the week. The format of this analysis included a question to the course's retrieval-augmented generated (RAG) LLM called [redacted for peer review] [12]. The RAG was grounded in VR-related academic publications, transcribed public talks, dedicated interviews and news articles about the course material, and various curriculum materials. The weekly topics that they were invited to query were pedagogically relevant to the topic of learning during that week. Students were prompted to query the LLM such that it provided a partially wrong answer. After receiving the LLM's response, students wrote a brief paragraph providing an improved answer.

Teaching and research teams were blind to the consenting status of students until the end of the course to ensure that all students received the same experience and treatment regardless of their consenting status. In line with the IRB protocol, students were all required to complete the self-report questionnaires for pedagogical purposes, and responses from non-consenting students were excluded from subsequent analyses. From the total 142 participants, participation rates varied across weeks (Week 1: 129, Week 2: 121, Week 3: 126, Week 4: 126, Week 5: 129, Week 6: 121). Participants identified as African, African-American or Black (N = 14), Asian or Asian-American (N = 58), Hispanic or Latinx (N = 12), Indigenous/Native American, Alaska Native, First Nations (N = 1), Middle Eastern (N = 4), Native Hawaiian or other Pacific Islander (N = 4), White (N = 30), more than one race (N = 7), decline to answer (N = 12). They were of ages 18-20 (N = 63), 21-24 (N = 68), 26 (N = 1) and declined to answer (N = 10) and were male (N = 57), female (N = 75) and declined to answer (N = 10).

AI literacy grouping. AI literacy was assessed using the short Meta-Artificial Intelligence Literacy Scale (MAILS) [13], administered in the 1st week of the class These items captured the ability to detect AI, apply AI to achieve goals, understand the advantages and disadvantages of AI, and its societal consequences (Cronbach's alpha = 0.74) (**Supplemental Table 6**). Students were classified as having high AI literacy if they scored above the sample median (N = 66) or having low AI literacy (N = 66) if they scored below the median. Those who did not answer the measure (N = 10) were excluded from the analysis.

Automated text analysis. We used Linguistic Inquiry and Word Count (LIWC) to create a psychological profile of each student using their language patterns . All student and LLM texts were scored using LIWC-22 [1], which maps word usage onto validated psychological and linguistic dimensions in its internal dictionary. Analyses evaluating linguistic features over time focused on style words, namely: first-person singular (I), impersonal pronouns (Ipron), third person singular (She/He), third person plural (They), first-person plural (We), and second person pronouns (You). LSM analyses focused on personal pronouns, impersonal pronouns, articles, conjunctions, auxiliary verbs, prepositions, adverbs, negations, and quantity words.

Statistical analysis. Linear mixed effects models were fitted separately for each LIWC feature using lme4 and lmerTest in R. To account for stable individual differences in language use, we parsed each linguistic variable into two components following an established person-mean centering procedure. The between-person component captures each participant's overall language usage across all weeks, while the within-person component captures week-to-week deviations from each participant's own mean. Additionally, lagged versions of each linguistic variable and their within-person centered counterparts were created to capture whether a participant's language use in a given week was predicted by their use in the previous week.

For the analyses of linguistic style matching, we used the following equation [8]:

$$LSM = 1 - \frac{|A - B|}{A + B + 0.0001}$$

where A and B represent the proportional usage of a given function word category for the interactants (LLM or human). A small constant (0.0001) was added to the denominator to prevent division by zero. Scores for each category were then averaged across all nine categories to produce a composite LSM score, with values approaching 1.0 indicating greater linguistic alignment (more "joint attention"). For the individual-level LSM analyses, linear mixed effects models were estimated using the lme4 and lmerTest packages in R. Each model accounted for the weekly LSM score on group (with students with high AI literacy as reference), week, and their interaction, with a random intercept for participants to account for repeated measures.

$$LSM_{it} = \beta_0 + \beta_1 (Group_i) + \beta_2 (Week_t) + \beta_3 (Group_i \ x \ Week_t) + u_i + \epsilon_{it}$$

For analysis of pronoun use over time, separate linear mixed effects models were estimated. Each model included the between-person mean, the lagged within-person deviation from mean, group (students with high AI literacy as reference), week, and a group-by-week interaction as fixed effects, with a random intercept for participant ID. The group-by-week interaction and within person differences was the primary term of interest, testing whether pronoun trajectories over time differed significantly across groups and within persons.

$$Y_{it} = \beta_0 + \beta_1 (Between\ Person\ Mean\ Differences_i) + \beta_2 (Within\ Person\ Lag_{it-1}) + \beta_3 (Group_i) + \beta_4 (Week_t) + \beta_5 (Group_i \ x \ Week_t) + u_i + \epsilon_{it}$$